Signals and Systems
Computer Homework 2

Seyed Kiarash Eshraghi

Professor: Dr. Arash Amini

Section 1

Part b

We defined a symbolic function for the projectile and used the two mentioned kinematic equations for it. Then, we used the `fplot` syntax to plot the trajectory.

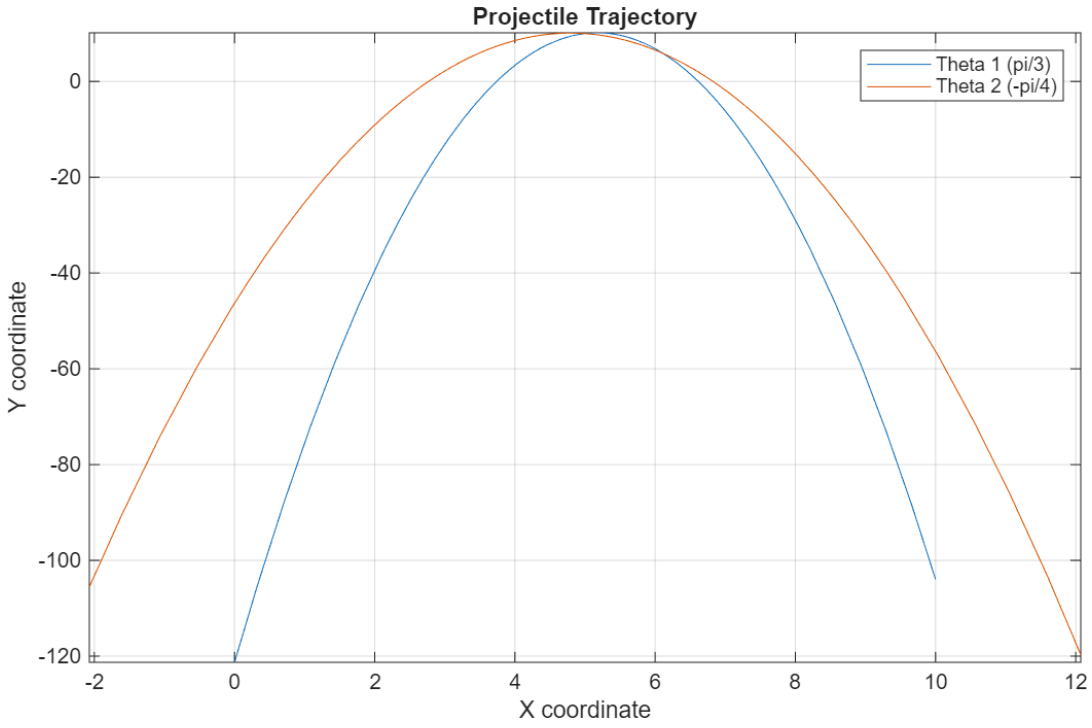


Figure 1: Projectile Trajectory

Part c

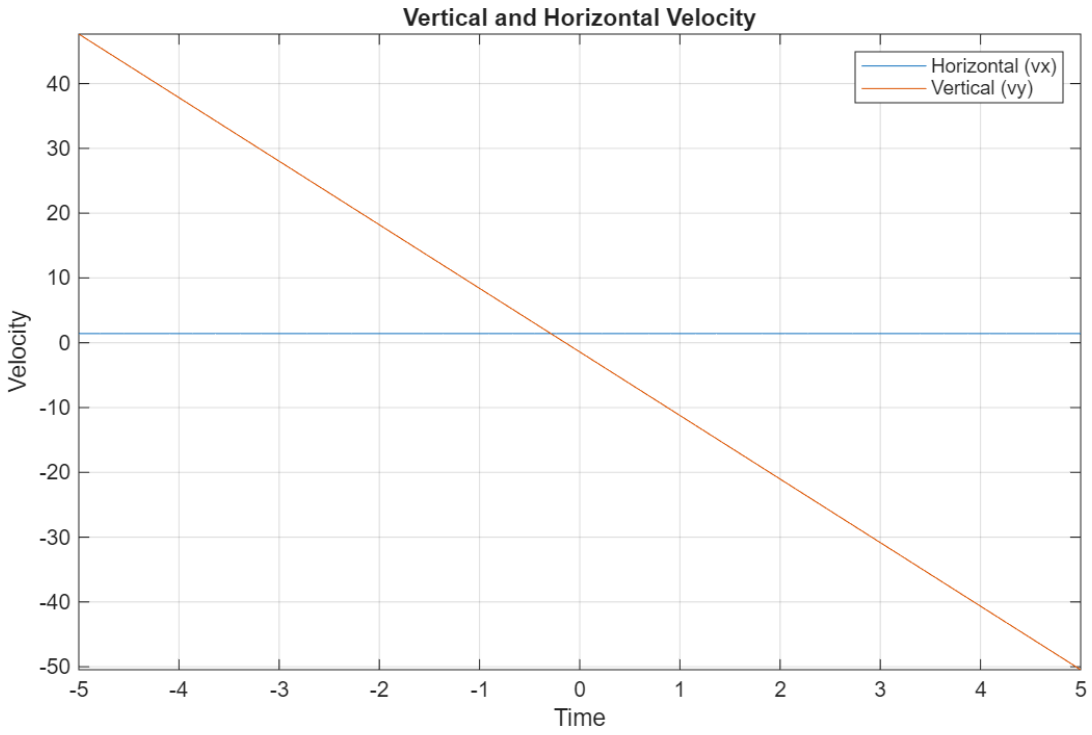


Figure 2: Vertical and Horizontal Velocity

For this part, I used the `diff` function (a symbolic function) to differentiate $y(t)$ and $x(t)$ with respect to time, and then I plotted them.

Section 2

Part a

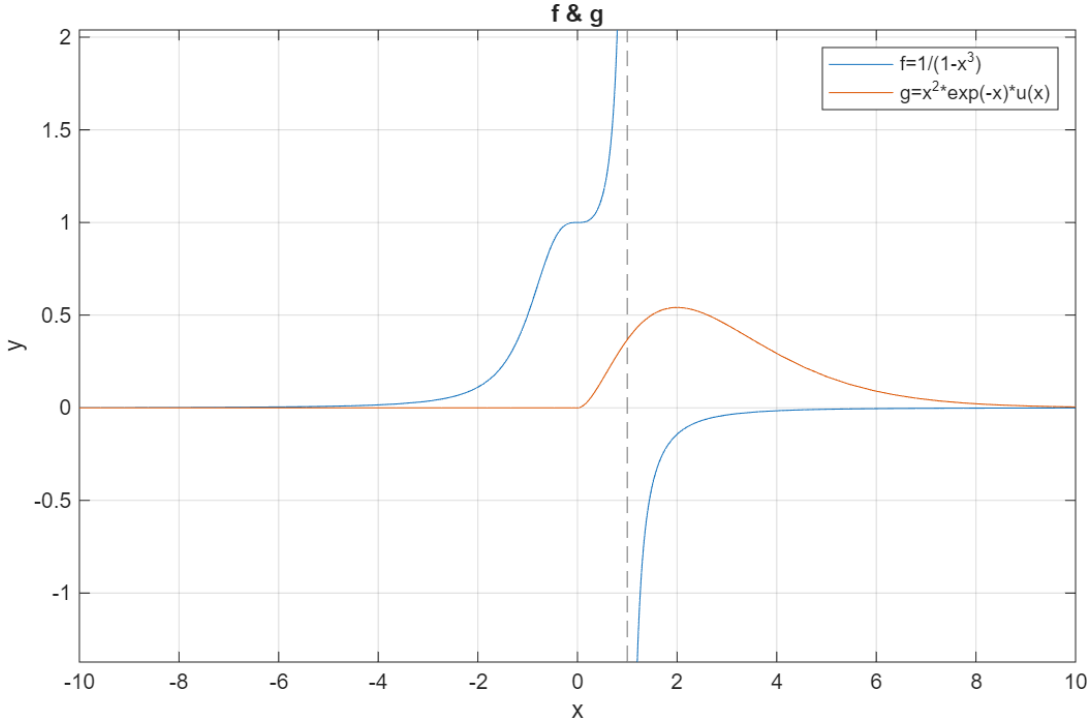


Figure 3: Functions f and g

Part b

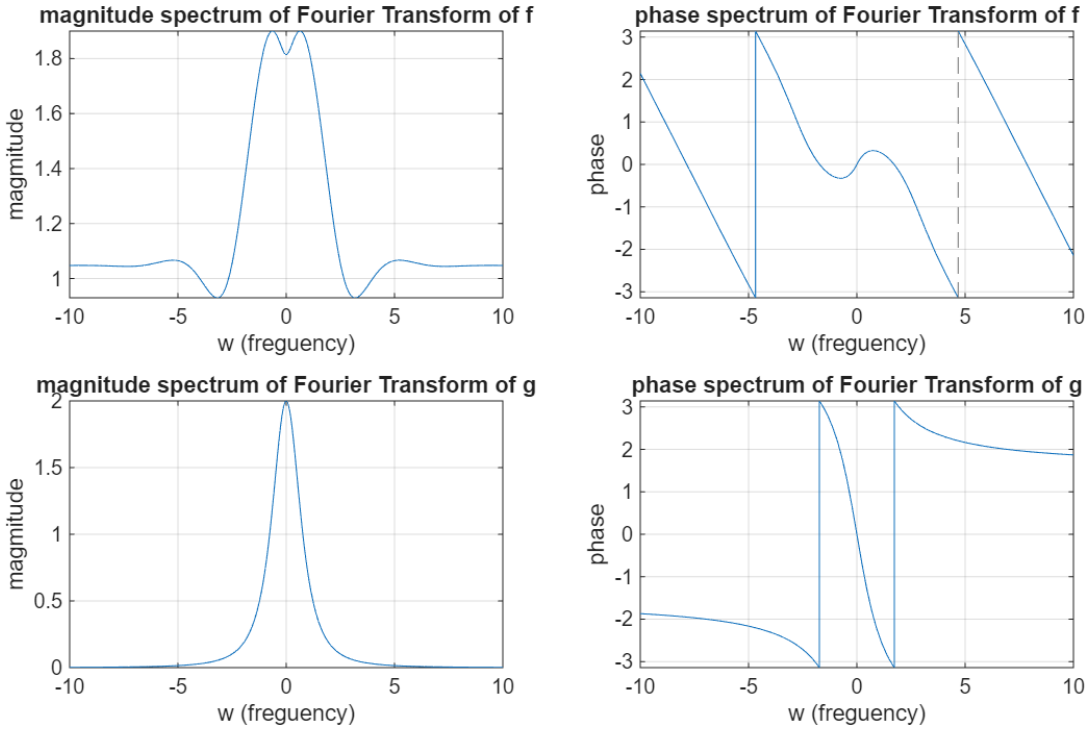


Figure 4: Magnitude and phase spectrum of Fourier Transform of f and g

Part c

Because of the $x = 1$ pole in the definition of $f(x)$, the MATLAB computing engine cannot handle the convolution of $f(x)$ and $\mathcal{F}\{f(x)\}(\omega)$. Therefore, we verified these relations for some discrete points.

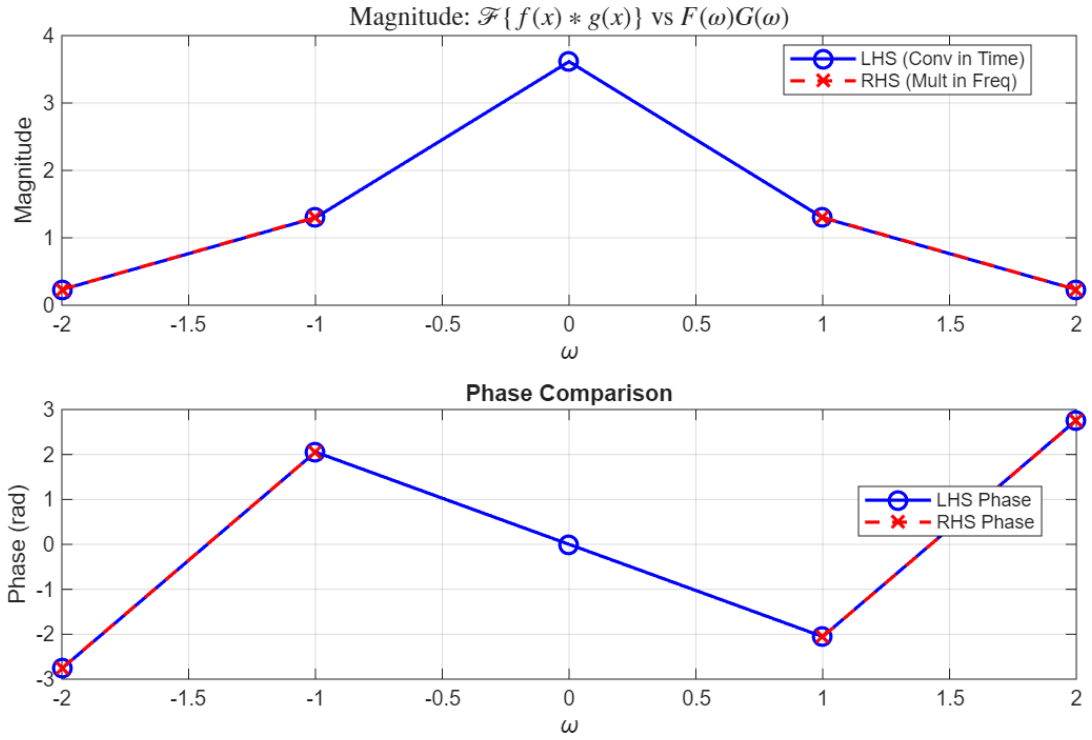


Figure 5: Magnitude and Phase Comparison (Convolution in Time)

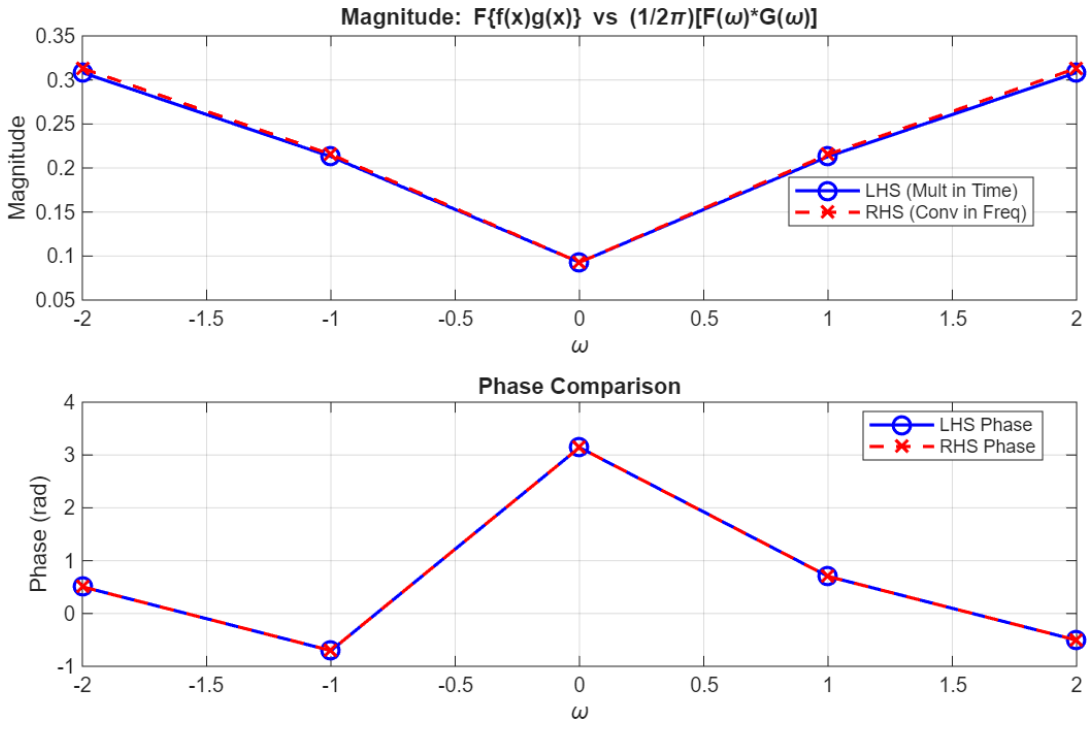


Figure 6: Magnitude and Phase Comparison (Multiplication in Time)

For $F(\omega) \times G(\omega)$, at zero frequency we have some functions like the Dirac delta function; therefore, we cannot plot it.

Section 3

Parts a & b

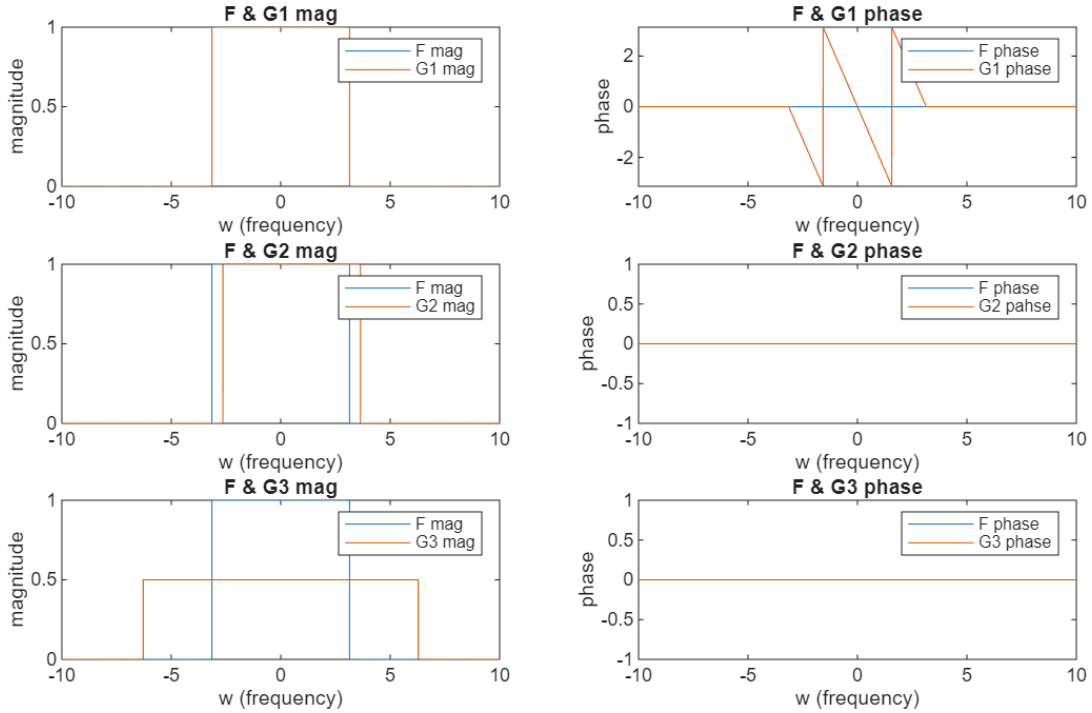


Figure 7: F and G1, G2, G3 magnitude and phase

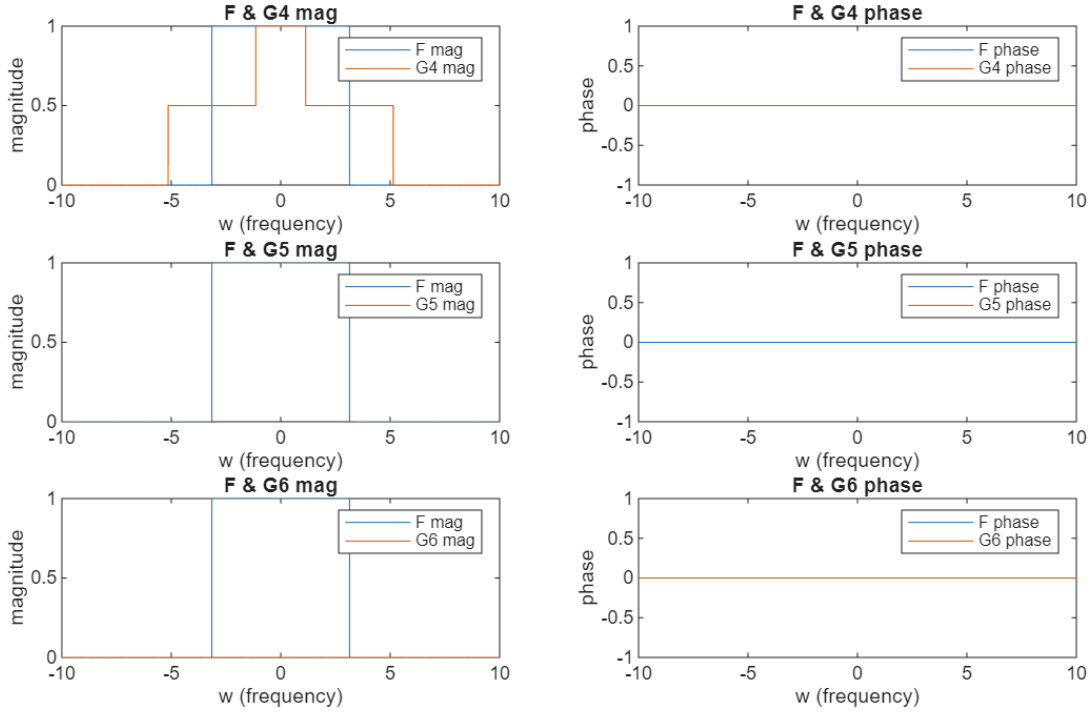


Figure 8: F and G4, G5, G6 magnitude and phase

As you can see, MATLAB didn't plot $G_5(\omega)$ because sinc(0) is undefined, and consequently, $g_5(0)$ is undefined by its formal definition.

Section 4

Part b

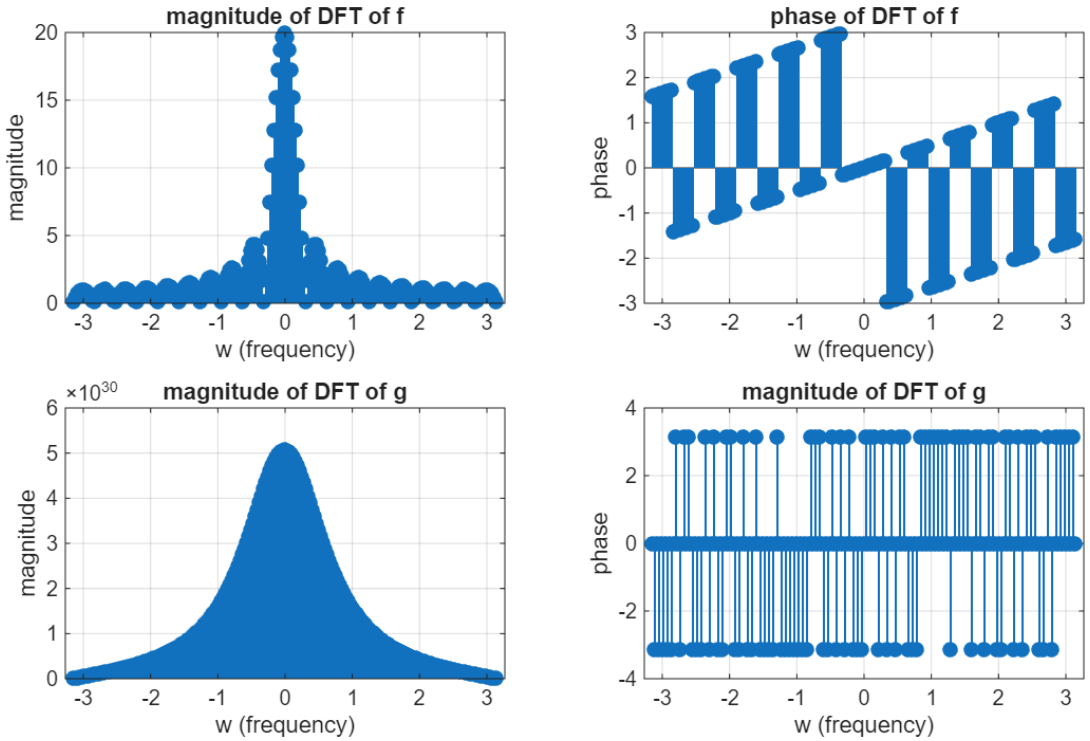


Figure 9: Magnitude and phase of DFT of f and g

Part c

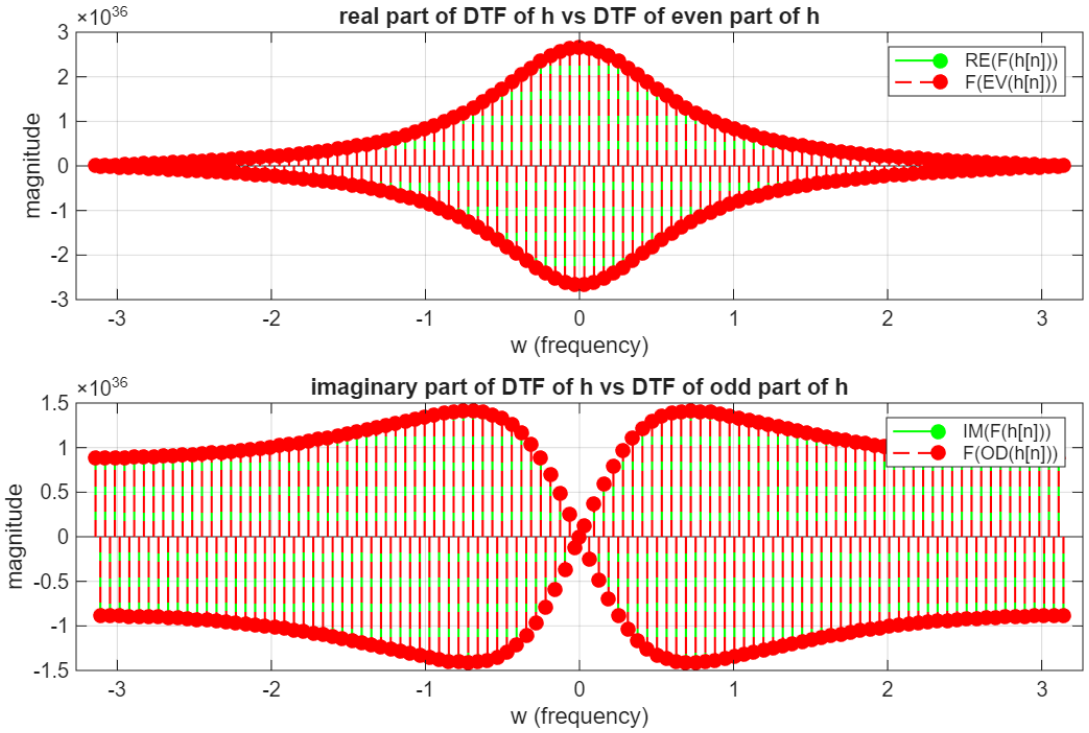


Figure 10: DTF of h vs DTF of even and odd parts of h